FISH++
MONITORING POLLUTION & POPULATION BELOW THE WATER
presented by : CPE11S4 GROUP 2

Di Prinzio et al. (2024) found that .....

- Treated wastewater discharge can increase pollution levels and negatively affect fish abundance, density, and biomass.
- Areas with higher pollution showed significant declines in fish populations, with the most polluted stream having no fish.

Raafat et al. (2023) found that .....

- Simulation models can be used to analyze pollution conditions and determine their possible effects on aquatic life.

As pollution levels increase, the conditions of their aquatic environment can become less suitable for their survival.

HENCE..

There is still a need for simple and accessible systems that can both monitor pollution levels and demonstrate their possible effects on aquatic life.

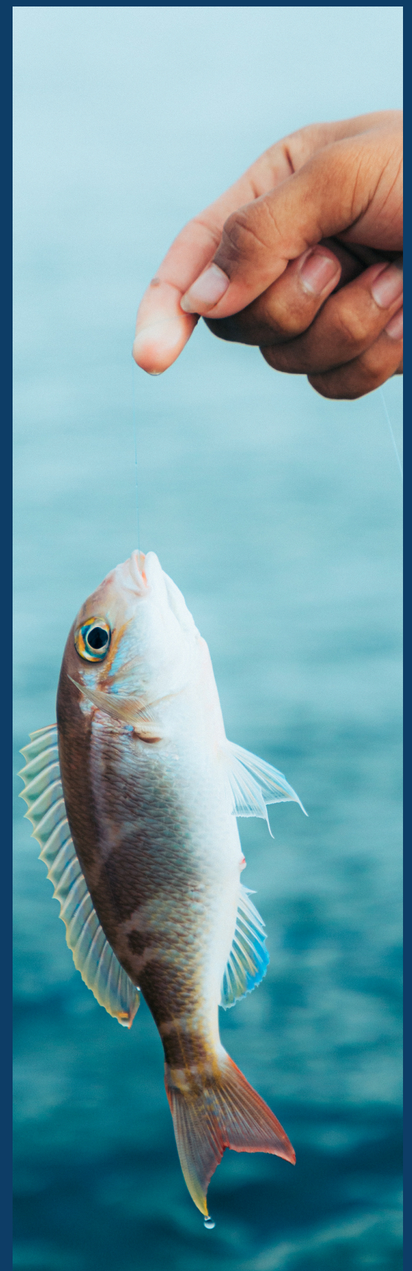
FISH POPULATION MONITORING AND SIMULATION PROGRAM (FISH++)

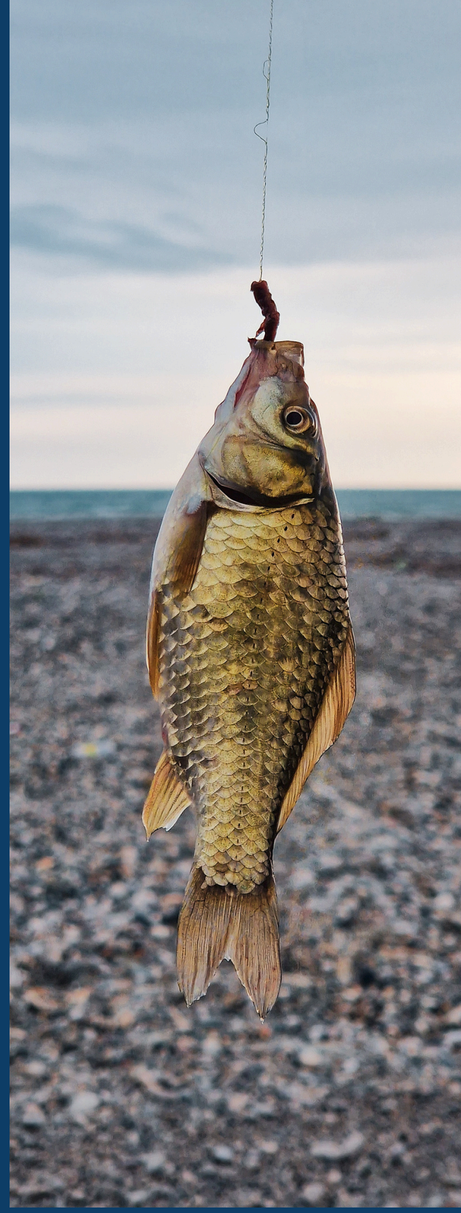
- Under SDG 14: Life Below Water
- Fish population monitoring component.
- Display relevant environmental information, such as the pollution level, water quality, and other factors that may influence the fish population.
- Simulation component.
- The program will use the environmental conditions associated with that area, particularly its pollution level.

WHY DO THIS?

POSSIBLE BENEFITS OF FISH++

- Early Detection
- Trend Monitoring
- Impact Visualization
- Environmental Awareness
- Decision Support





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- Raafat, P. B., Ibrahim, F. N., & Saleh, A. (2023). On determining conditions and suitable locations for fish survival by using the solution of the two coupled pollution and aeration equations. *Scientific Reports*, 13, 6560. https://doi.org/10.1038/s41598-023-33368-9

- https://sdgs.un.org/goals/goal14

THANK YOU
FOR LISTENING !!